CSA6303

Threat Intelligence and Network Security

AT1: Application-Based Questions

Answer Sheet

Topics: Network Security Fundamentals | Security Goals (CIA Triad) | Types of Network Threats | Introduction to Threat Intelligence | Types of Threat Intelligence (Strategic, Tactical, Operational) | Threat Intelligence Lifecycle | Applications in IT Industry

CO1: Interpret security principles, network threats, and the role of threat intelligence in cybersecurity.

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Total Marks	50
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Code of Conduct

I T. Raju (192465038) (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: _____

CSA6303 - Threat Intelligence and Network Security

AT1: Application-Based Questions - Answers (CO1)

Q1. CIA Triad - Scenario Identification (5 marks)

Principle violated: Integrity.

Integrity means data stays accurate and is changed only by authorized parties through authorized means. The attacker used illegitimate database write access to silently alter product prices, so the stored data is no longer trustworthy or correct.

Availability is NOT violated: the site stayed fully online and no downtime occurred, so legitimate users could still reach the service.

Confidentiality: not demonstrated by the facts given (only modification is described). However, since an attacker gained unauthorized access to the database, a confidentiality breach is possible and should be investigated.

Conclusion: Integrity is the principle clearly and directly violated.

Q2. Annualized Loss Expectancy (10 marks)

Given: Asset Value (AV) = $50,000; Exposure Factor (EF) = 40% = 0.40; Annualized Rate of Occurrence (ARO) = 2.

> **Formulas:** SLE = AV x EF | ALE = SLE x ARO

Step 1 - SLE = 50,000 x 0.40 = **$20,000** per incident

Step 2 - ALE = 20,000 x 2 = **$40,000** per year

Interpretation: The company can expect to lose about $40,000 a year from this threat. Any control costing less than $40,000 a year that fully removes the risk is cost-justified; this is the basis for comparing safeguard cost against ALE reduction.

Q3. Brute-Force Attack Time Estimation (10 marks)

Given: length n = 8; character set size c = 62 (26 + 26 + 10); rate = 1×10^9 guesses/second.

> **Keyspace** = $c^n = 62^8$ = **218,340,105,584,896** (approx. 2.18×10^{14}) combinations

Worst-case time (attacker tries every combination, correct password is last):

> Time (s) = keyspace / rate = $2.18340105584896 \times 10^{14} / 10^9$ = 218,340.1 seconds
>
> Time (hours) = 218,340.1 / 3,600 = **approx. 60.65 hours** (about 2.53 days)

On average the password is found after searching half the keyspace, about 30.3 hours. Each extra character multiplies the time by 62, which is why longer passwords and slow hashing matter.

Q4. Risk Prioritization Matrix (10 marks)

> **Formula:** Risk Score = Likelihood (L) x Impact (I)

Threat	L	I	Risk Score (L x I)	Priority Rank
(a) Ransomware	4	5	4 x 5 = 20	1
(e) Unpatched server exploit	4	4	4 x 4 = 16	2

Threat	L	I	Risk Score (L x I)	Priority Rank
(b) Phishing	5	3	5 x 3 = 15	3
(c) Insider data theft	2	5	2 x 5 = 10	4
(d) DDoS	3	3	3 x 3 = 9	5

Priority order: Ransomware (20) > Unpatched server exploit (16) > Phishing (15) > Insider data theft (10) > DDoS (9).

Remarks: Patching and budget go first to ransomware (backups, EDR, segmentation) and to unpatched servers (patch management). Phishing, ranked third, is closely linked to ransomware delivery, so awareness training and email filtering also reduce the top risk. Insider theft has high impact but low likelihood, so it is handled by access control and monitoring.

Q5. Passive vs. Active Threat Classification (5 marks)

Scenario	Type	Primary CIA principle threatened	Reason
(a) Sniffing unencrypted traffic on public Wi-Fi	Passive	Confidentiality	Attacker only eavesdrops and reads data; nothing is altered, so it is hard to detect.
(b) ARP spoofing to redirect traffic	Active	Integrity (also enables confidentiality loss and disruption)	Attacker injects forged ARP replies, altering network mappings and traffic flow (man-in-the-middle), and can modify data in transit.
(c) Long-term reconnaissance without altering anything	Passive	Confidentiality	Attacker only gathers information about the network and systems, with no modification of data or systems.

Q6. Threat Intelligence Lifecycle Sequencing (10 marks)

Order	Phase	Step
1	Planning and Direction (Requirements)	(iii) The request is logged as an intelligence requirement.
2	Collection	(iv) OSINT and threat feeds are pulled for the group's known TTPs.
3	Processing and Analysis	(i) Analysts correlate collected IOCs with internal logs to judge exposure.
4	Dissemination	(ii) A report is sent to the CISO.
5	Feedback	(v) The CISO's follow-up questions shape the next collection cycle.

Correct sequence: (iii) → (iv) → (i) → (ii) → (v)

The cycle is continuous: feedback in phase 5 feeds back into Planning and Direction, starting a refined next iteration.